RAJALAKSHMI
ENGINEERING
COLLEGE

G Pranav Iyer 251801233-2025-AIDS G2

Started on	Monday, 24 August 2026, 3:13 PM
State	Finished
Completed on	Monday, 24 August 2026, 3:18 PM
Time taken	5 mins 12 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00

```
Convert the following algorithm into a program and find its time
```

```
complexity using counter method.

void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, i, j, k, c = 2;

    scanf("%d", &n);

    for (i = n / 2; i < n; i++)
    {
        c += 2;

        for (j = 1; j < n; j = 2 * j)
        {
            c += 2;

            for (k = 1; k < n; k = k * 2)
            {
                c += 2;
            }
        }
    }

    printf("%d", c);

    return 0;
}
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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